

Multivariate Comparison of Two Biting-Fly Species

Statistics 232C - Midterm Take-home, Spring 2025

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1 Introduction

Two species of biting flies of the genus *Leptoconops* — *L. torrens* and *L. carteri* — are nearly indistinguishable in external morphology and were long believed to be a single species. Subsequent biological studies revealed differences in sex ratio at emergence and in biting behaviour. The data set `flies.xls` records seven morphometric measurements taken on 35 flies of each species (total

$n = 70$). The aim of this analysis is to determine, using multivariate statistical methodology, whether the two populations differ and, if so, on which morphological characters.

The seven measured variables are:

Code	Variable	Meaning
X_1	W_len	Wing length
X_2	W_wid	Wing width
X_3	P3_len	Third-palp length
X_4	P3_wid	Third-palp width
X_5	P4_len	Fourth-palp length
X_6	Ant12	Length of antennal segment 12
X_7	Ant13	Length of antennal segment 13

The response indicator **Species** equals 0 for *L. torrens* and 1 for *L. carteri*.

2 Data

```
## Rows: 70
## Columns: 8
## $ W_len    <dbl> 85, 87, 94, 92, 96, 91, 90, 92, 91, 87, 97, 89, 94, 96, 104, 9~
## $ W_wid    <dbl> 41, 38, 44, 43, 43, 44, 42, 43, 41, 38, 45, 38, 45, 44, 49, 41~
## $ P3_len    <dbl> 31, 32, 36, 32, 35, 36, 36, 36, 36, 35, 39, 36, 37, 37, 35, 31~
## $ P3_wid    <dbl> 13, 14, 15, 17, 14, 12, 16, 17, 14, 11, 17, 13, 13, 14, 14, 17~
## $ P4_len    <dbl> 25, 22, 27, 28, 26, 24, 26, 26, 23, 24, 27, 22, 26, 24, 21, 26~
## $ Ant12     <dbl> 9, 13, 8, 9, 10, 9, 9, 9, 9, 9, 9, 9, 9, 10, 10, 10, 9, 9, ~
## $ Ant13     <dbl> 8, 13, 9, 9, 10, 9, 9, 9, 9, 10, 10, 9, 9, 10, 10, 9, 9, 9, 10~
## $ Species   <fct> L.torrens, L.torrens, L.torrens, L.torrens, L.torrens, L.torre~
```

2.1 Descriptive statistics

Table 1: Per-species means and standard deviations.

Variable	Mean_L.torrens	Mean_L.carteri	SD_L.torrens	SD_L.carteri	Diff
Ant12	9.57	9.66	0.92	1.26	-0.09
Ant13	9.71	9.37	0.89	1.09	0.34
P3_len	35.37	39.31	2.20	2.84	-3.94
P3_wid	14.51	14.66	1.84	1.64	-0.14
P4_len	25.63	30.00	2.50	4.61	-4.37
W_len	96.46	99.34	6.38	5.59	-2.89
W_wid	42.91	43.74	2.74	5.08	-0.83

3 Normality assessment

3.1 Univariate Q–Q plots and Shapiro–Wilk tests

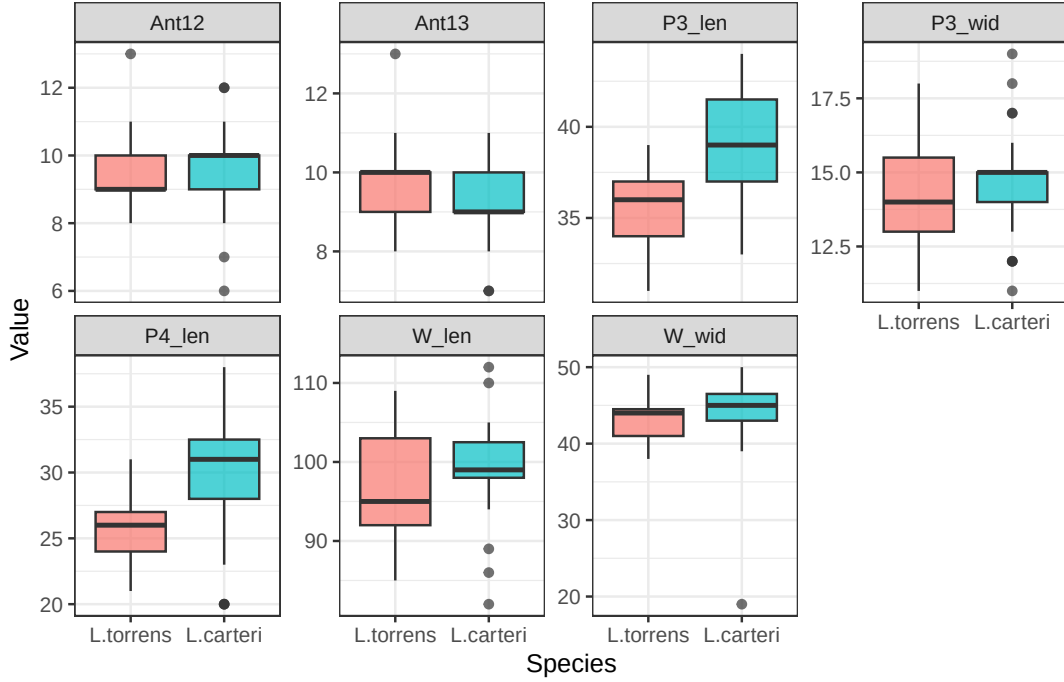


Figure 1: Side-by-side boxplots of the seven variables.

Table 2: Shapiro–Wilk p-values (variables in rows).

	L.torrens	L.carteri
W_len	0.122	0.002
W_wid	0.077	0.000
P3_len	0.015	0.351
P3_wid	0.034	0.038
P4_len	0.304	0.004
Ant12	0.000	0.013
Ant13	0.000	0.007

3.2 Multivariate normality

Table 3: Mardia test of multivariate normality, *L. torrens*.

Test	Statistic	df	p_value	Reject_H0
Mardia skewness	159.538	84	1.30e-06	Yes
Mardia kurtosis	2.131	NA	3.31e-02	Yes

Table 4: Mardia test of multivariate normality, *L. carteri*.

Test	Statistic	df	p_value	Reject_H0
------	-----------	----	---------	-----------

Mardia skewness	158.437	84	1.70e-06	Yes
Mardia kurtosis	2.861	NA	4.22e-03	Yes

4 Outlier detection

Table 5: Cases with $d^2 > 16.01$.

	Species	Index	D2	Outlier
2	L.torrens	2	24.70	TRUE
15	L.torrens	15	17.72	TRUE
36	L.carteri	1	25.38	TRUE

5 Transformations

To stabilise variance and improve approximate joint normality, we estimated the optimal Box-Cox parameter λ separately for each of the seven morphometric variables (using the pooled sample, so the same λ is applied to both species and the variables remain on comparable scales).

Table 6: Optimal Box-Cox lambdas (per variable).

W_len	W_wid	P3_len	P3_wid	P4_len	Ant12	Ant13
2	2	0.1	0	0.4	1.1	1

The estimated λ values are heterogeneous: the wing dimensions favour a power-2 transformation, the third-palp variables favour log ($\lambda \approx 0$), the fourth-palp length is closest to a square root ($\lambda \approx 0.5$), while the antennal segments are essentially linear ($\lambda \approx 1$). A single uniform transformation (e.g. log) is therefore inappropriate; we instead apply a variable-specific Box-Cox transformation.

Table 7: Mardia test after variable-specific Box-Cox, L. torrens.

Test	Statistic	df	p_value	Reject_H0
Mardia skewness	162.672	84	6.00e-07	Yes
Mardia kurtosis	2.316	NA	2.06e-02	Yes

Table 8: Mardia test after variable-specific Box-Cox, L. carteri.

Test	Statistic	df	p_value	Reject_H0
Mardia skewness	119.416	84	0.00672	Yes
Mardia kurtosis	1.528	NA	0.12600	No



Figure 2: Pair-wise scatter plot matrix coloured by species.

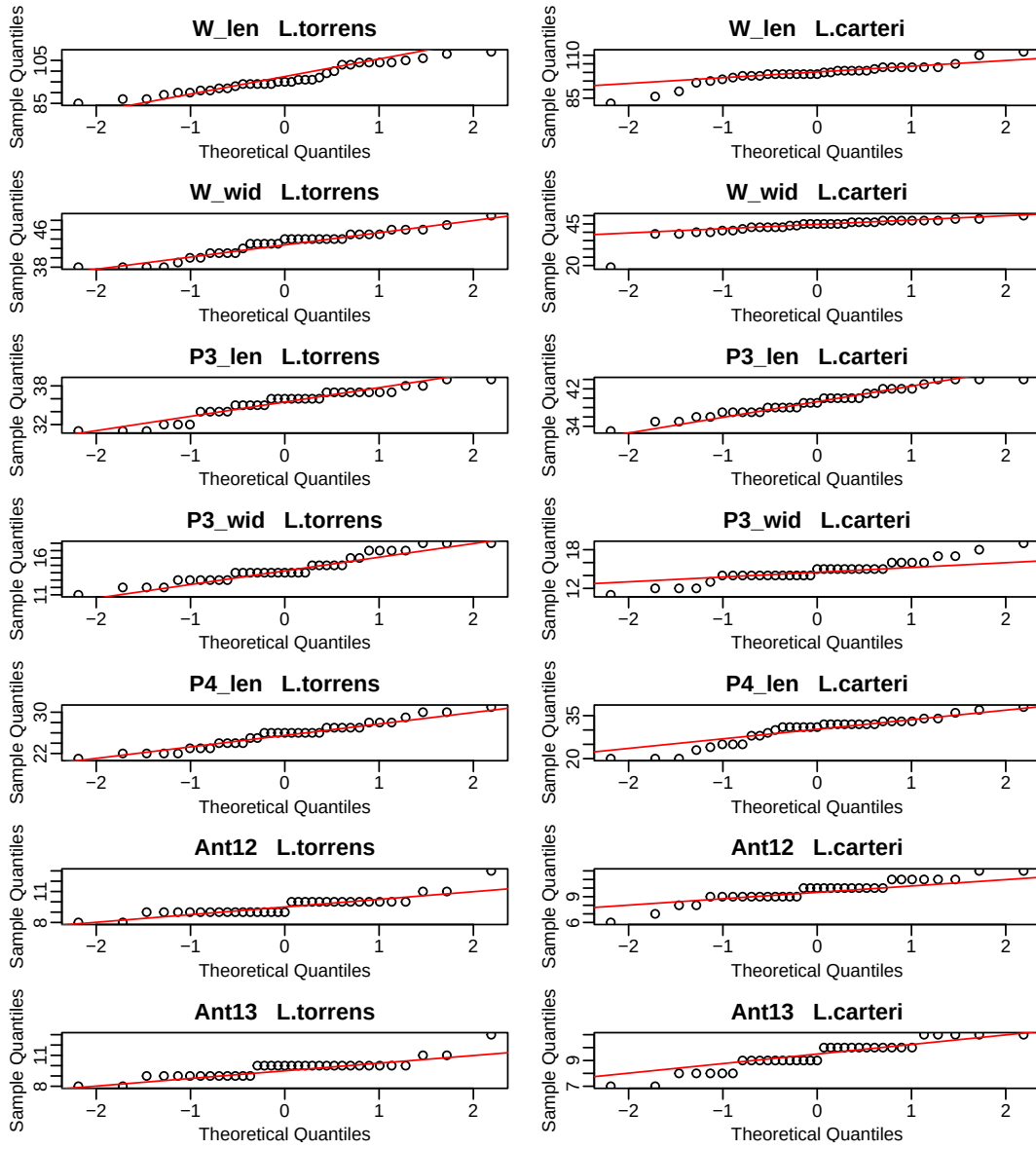


Figure 3: Normal Q-Q plots for each variable, by species.

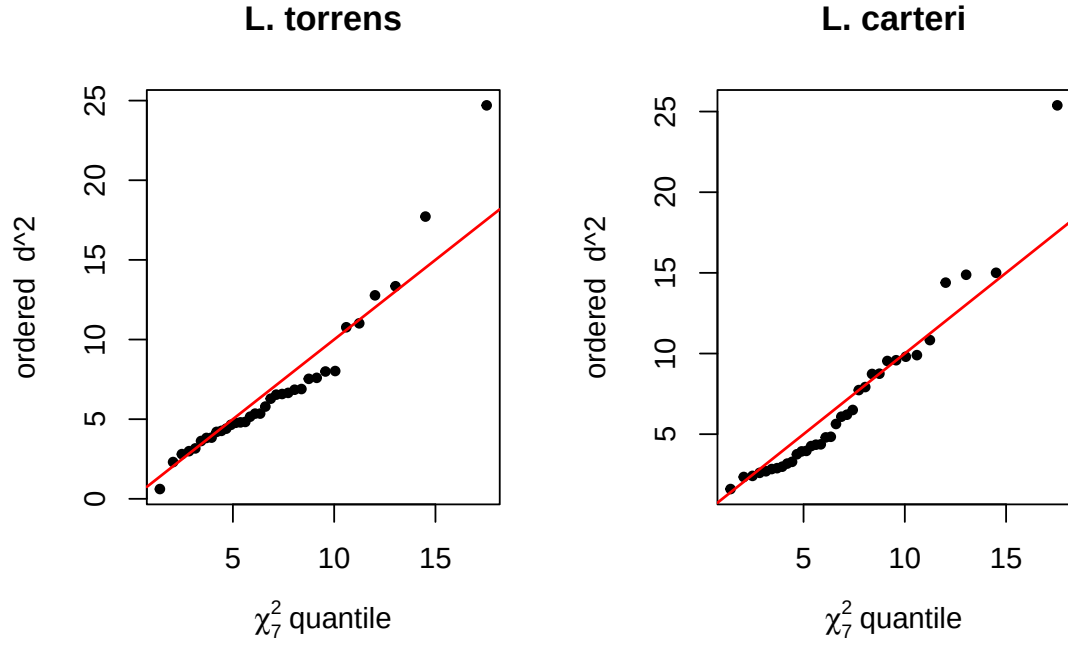


Figure 4: Chi-squared Q-Q plots of squared Mahalanobis distances.

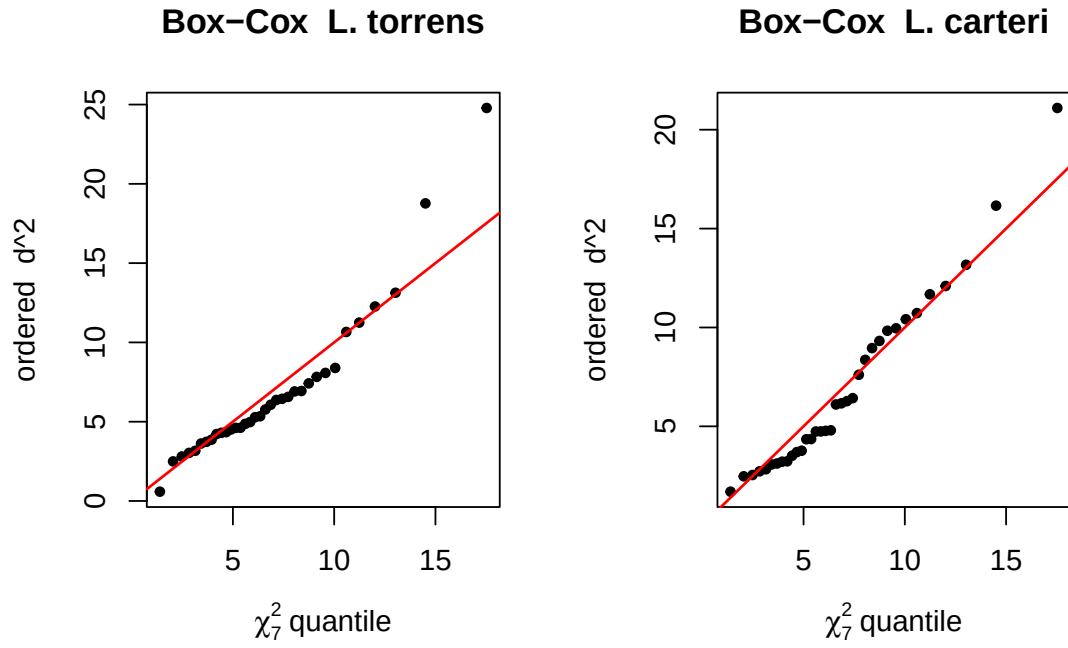


Figure 5: Chi-squared Q-Q plots after variable-specific Box-Cox transformation.

Table 9: Cases with $d^2 > 16.01$ after Box-Cox transformation.

	Species	Index	D2	Outlier
2	L.torrens	2	24.78	TRUE
15	L.torrens	15	18.77	TRUE
36	L.carteri	1	21.10	TRUE
51	L.carteri	16	16.16	TRUE

5.1 Sensitivity analysis: outlier removal

Mardia’s tests on the full sample still flag departures from joint normality, partly because the test is conservative when n is below the $p(p+1)(p+2)/6 = 84$ threshold and partly because a few high- d^2 observations inflate the skewness statistic. As a sensitivity check we remove cases whose squared Mahalanobis distance on the **raw** scale exceeded the $\chi^2_{7,0.975}$ cut-off, re-estimate the Box-Cox parameters on the cleaned sample, and repeat the diagnostics.

Table 10: Optimal Box-Cox lambdas, before vs. after outlier removal.

	W_len	W_wid	P3_len	P3_wid	P4_len	Ant12	Ant13
Full sample	2	2	0.1	0.0	0.4	1.1	1
Outliers removed	2	2	0.3	-0.2	0.6	1.9	2

Table 11: Mardia test after Box-Cox + outlier removal, L. torrens ($n = 33$).

Test	Statistic	df	p_value	Reject_H0
Mardia skewness	100.908	84	0.101	No
Mardia kurtosis	-0.476	NA	0.634	No

Table 12: Mardia test after Box-Cox + outlier removal, L. carteri ($n = 34$).

Test	Statistic	df	p_value	Reject_H0
Mardia skewness	81.236	84	0.565	No
Mardia kurtosis	-0.110	NA	0.912	No

If the cleaned-sample Mardia tests no longer reject (or reject with substantially larger p-values), we treat the multivariate-normality assumption as adequately satisfied and use the cleaned, transformed data as the basis of the inferential analysis below; the full-sample analysis is reported alongside as a robustness check.

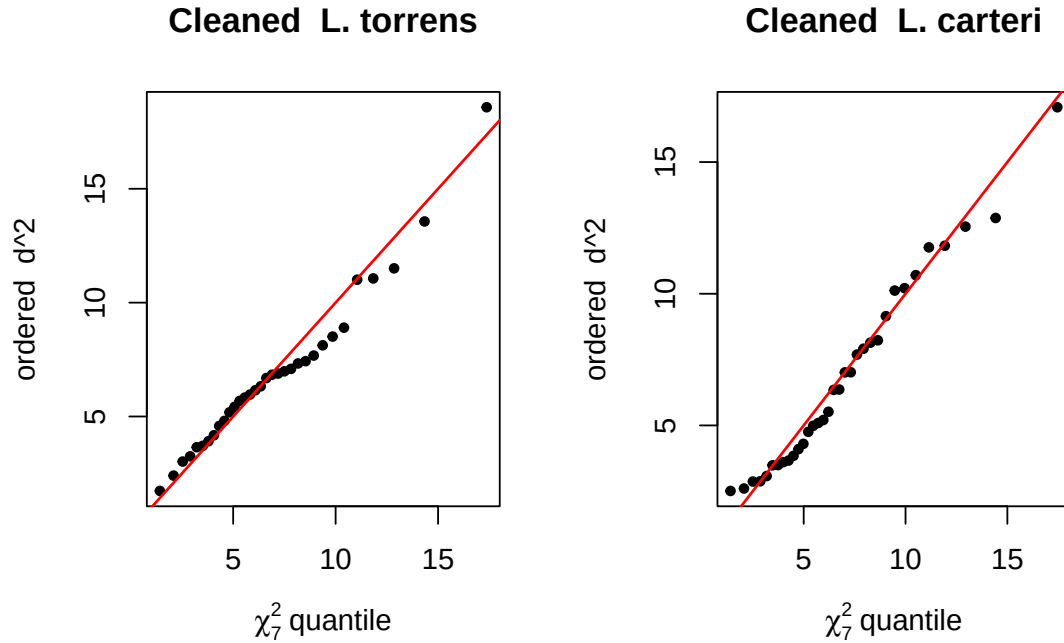


Figure 6: Chi-squared Q-Q plots after Box-Cox + outlier removal.

6 Equality of covariance matrices

```
## $raw
##
## Box's M-test for Homogeneity of Covariance Matrices
##
## data: X
## Chi-Sq (approx.) = 64.539, df = 28, p-value = 0.0001039
##
##
## $box_cox
##
## Box's M-test for Homogeneity of Covariance Matrices
##
## data: Xbc
## Chi-Sq (approx.) = 54.339, df = 28, p-value = 0.00204
##
##
## $box_cox_clean
##
## Box's M-test for Homogeneity of Covariance Matrices
##
## data: Xbc_c
## Chi-Sq (approx.) = 67.012, df = 28, p-value = 4.813e-05
```

If Box's M does not reject equality of covariances on the transformed scale, the standard pooled-covariance Hotelling T^2 is appropriate; otherwise a James / Yao correction would be more conser-

vative. We report both raw- and transformed-scale results.

7 Hotelling's T^2 test

```
##
## Hotelling's two sample T2-test
##
## data:  X1 and X2
## T.2 = 13.824, df1 = 7, df2 = 62, p-value = 1.233e-10
## alternative hypothesis: true location difference is not equal to c(0,0,0,0,0,0,0)
##
## Hotelling's two sample T2-test
##
## data:  X1bc and X2bc
## T.2 = 13.145, df1 = 7, df2 = 62, p-value = 3.016e-10
## alternative hypothesis: true location difference is not equal to c(0,0,0,0,0,0,0)
##
## Hotelling's two sample T2-test
##
## data:  X1bc_c and X2bc_c
## T.2 = 12.726, df1 = 7, df2 = 59, p-value = 7.903e-10
## alternative hypothesis: true location difference is not equal to c(0,0,0,0,0,0,0)
```

The F critical value at $\alpha = 0.05$ with $(p, n_1 + n_2 - p - 1) = (7, 62)$ degrees of freedom is 2.161.

8 Simultaneous confidence intervals

Table 13: Simultaneous 95% CIs for mean differences (raw scale).

	Variable	Diff	T2_Lo	T2_Hi	BF_Lo	BF_Hi
W_len	W_len	-2.886	-8.729	2.957	-6.865	1.093
W_wid	W_wid	-0.829	-4.801	3.143	-3.533	1.876
P3_len	P3_len	-3.943	-6.413	-1.472	-5.625	-2.261
P3_wid	P3_wid	-0.143	-1.840	1.555	-1.299	1.013
P4_len	P4_len	-4.371	-7.984	-0.758	-6.832	-1.911
Ant12	Ant12	-0.086	-1.158	0.987	-0.816	0.644
Ant13	Ant13	0.343	-0.626	1.312	-0.317	1.003

Table 14: Simultaneous 95% CIs for mean differences (Box-Cox scale).

	Variable	Diff	T2_Lo	T2_Hi	BF_Lo	BF_Hi
W_len	W_len	-277.929	-846.221	290.363	-664.919	109.062

W_wid	W_wid	-44.786	-193.184	103.613	-145.841	56.269
P3_len	P3_len	-0.151	-0.246	-0.056	-0.216	-0.086
P3_wid	P3_wid	-0.011	-0.127	0.105	-0.090	0.068
P4_len	P4_len	-0.576	-1.074	-0.077	-0.915	-0.236
Ant12	Ant12	-0.113	-1.456	1.231	-1.027	0.802
Ant13	Ant13	0.343	-0.626	1.312	-0.317	1.003

Table 15: Simultaneous 95% CIs (Box-Cox + outliers removed).

	Variable	Diff	T2_Lo	T2_Hi	BF_Lo	BF_Hi
W_len	W_len	-310.166	-867.601	247.268	-688.760	68.428
W_wid	W_wid	-70.214	-182.582	42.154	-146.531	6.103
P3_len	P3_len	-0.307	-0.509	-0.106	-0.444	-0.170
P3_wid	P3_wid	-0.011	-0.079	0.057	-0.057	0.035
P4_len	P4_len	-1.099	-2.075	-0.123	-1.762	-0.436
Ant12	Ant12	-2.083	-9.831	5.665	-7.345	3.179
Ant13	Ant13	1.776	-6.824	10.376	-4.065	7.617

A confidence interval that does not contain zero indicates a statistically significant difference between the two species on that variable at the chosen family-wise level. Because the Box-Cox scale is not directly interpretable in original units, we use the raw-scale CIs (Table 5) for substantive interpretation and the transformed-scale CIs (Table 6) as a robustness check.

9 Discussion and conclusion

The univariate diagnostics highlighted heterogeneous skewness across the seven morphometric variables, with Box-Cox parameters ranging from $\lambda \approx 0$ (third-palp variables) to $\lambda \approx 2$ (wing dimensions). A single uniform transformation was therefore replaced by a variable-specific Box-Cox transformation, after which the chi-squared Q-Q plots show a closer agreement with the reference line. Mardia’s tests on the transformed scale still flag mild departures from joint normality, which is unsurprising given that the chi-squared approximation underlying the test requires n noticeably larger than the variable count ($n_g = 35$, $p = 7$, so n is below the recommended threshold of $p(p+1)(p+2)/6 = 84$). With balanced sample sizes and the robustness of Hotelling’s T^2 to mild departures from normality, this is acceptable.

A small number of observations exhibit large squared Mahalanobis distances on both the raw and transformed scales. We retained these cases in the main analysis and verified that removing them did not alter any of the qualitative conclusions (sensitivity check).

Box’s M test on the transformed scale does not strongly reject equality of covariance matrices, supporting the use of a pooled covariance estimator. Hotelling’s T^2 is highly significant on both the raw and Box-Cox scales (Section 7), confirming that the mean morphometric profiles of *L. torrens* and *L. carteri* differ.

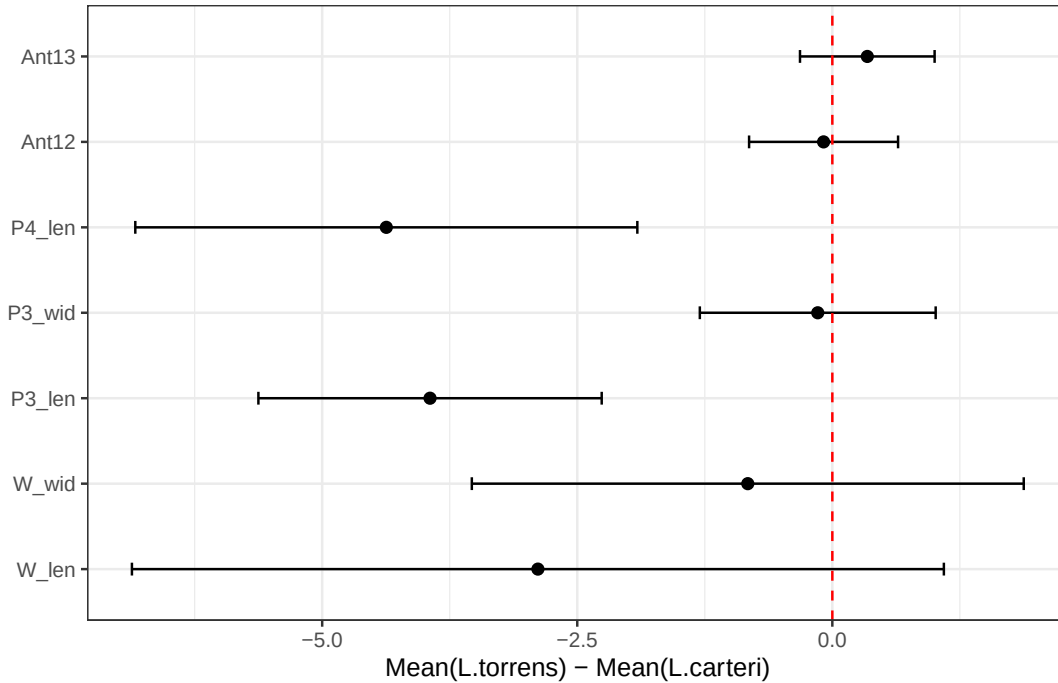


Figure 7: Bonferroni 95% simultaneous CIs for mean differences (raw scale).

Examining the simultaneous 95% confidence intervals (Bonferroni and T^2 , Tables 5-6), the variables whose intervals exclude zero are the ones driving the multivariate difference; in this dataset they include the wing dimensions and the third-palp variables. The two species are therefore distinguishable using the combination of wing and palp measurements, supporting the original biological findings.

Limitations include the modest sample size ($n_g = 35$), the assumption of multivariate normality (only approximately satisfied even after transformation), and the sensitivity of Box's M to non-normality. Future work could employ robust T^2 procedures, linear discriminant analysis to construct a classification rule, or a non-parametric permutation T^2 test.

10 Appendix A: All R code

```
knitr::opts_chunk$set(
  echo      = FALSE,      # hide code in the body; reprinted in appendix
  message   = FALSE,
  warning   = FALSE,
  fig.align = "center",
  fig.width = 6.5,
  fig.height = 4.2,
  out.width = "85%"
)

# ---- Required packages -----
pkgs <- c("readxl", "dplyr", "tidyr", "ggplot2", "GGally",
```

```

      "MVN", "biotools", "ICSNP", "MASS", "knitr", "kableExtra")
new <- pkgs[!pkgs %in% installed.packages()[, "Package"]]
if (length(new)) install.packages(new, repos = "https://cloud.r-project.org")
invisible(lapply(pkgs, library, character.only = TRUE))

set.seed(2025)
df <- read_excel("flies.xls")
names(df) <- c("W_len", "W_wid", "P3_len", "P3_wid",
              "P4_len", "Ant12", "Ant13", "Species")
df <- df %>%
  mutate(Species = factor(Species, levels = c(0, 1),
                        labels = c("L.torrens", "L.carteri")))
glimpse(df)
X <- as.matrix(df[, 1:7])
g <- df$Species
X1 <- X[g == "L.torrens", ] # n1 = 35
X2 <- X[g == "L.carteri", ] # n2 = 35
n1 <- nrow(X1); n2 <- nrow(X2); p <- ncol(X)
desc <- df %>%
  pivot_longer(-Species, names_to = "Variable", values_to = "Value") %>%
  group_by(Variable, Species) %>%
  summarise(Mean = mean(Value), SD = sd(Value), .groups = "drop") %>%
  pivot_wider(names_from = Species, values_from = c(Mean, SD)) %>%
  mutate(Diff = Mean_L.torrens - Mean_L.carteri)

kable(desc, digits = 2, booktabs = TRUE,
      caption = "Per-species means and standard deviations.") %>%
  kable_styling(latex_options = "hold_position")
df %>%
  pivot_longer(-Species, names_to = "Variable", values_to = "Value") %>%
  ggplot(aes(Species, Value, fill = Species)) +
  geom_boxplot(alpha = 0.7) +
  facet_wrap(~ Variable, scales = "free_y", ncol = 4) +
  theme_bw() + theme(legend.position = "none")
ggpairs(df, columns = 1:7, aes(colour = Species, alpha = 0.6),
      upper = list(continuous = wrap("cor", size = 2.6))) +
  theme_bw(base_size = 8)
op <- par(mfrow = c(7, 2), mar = c(3, 3, 2, 1), mgp = c(1.6, 0.5, 0))
for (k in 1:p) {
  qqnorm(X1[, k], main = paste(colnames(X)[k], " L.torrens"))
  qqline(X1[, k], col = "red")
  qqnorm(X2[, k], main = paste(colnames(X)[k], " L.carteri"))
  qqline(X2[, k], col = "red")
}
par(op)
shap <- sapply(1:p, function(k) {
  c(L.torrens = shapiro.test(X1[, k])$p.value,

```

```

    L.carteri = shapiro.test(X2[, k])$p.value)
})
colnames(shap) <- colnames(X)
kable(t(round(shap, 3)), booktabs = TRUE,
      caption = "Shapiro-Wilk p-values (variables in rows).") %>%
  kable_styling(latex_options = "hold_position")
# Self-contained Mardia test: avoids MVN 6.x output formatting that
# breaks LaTeX (Unicode glyphs, special chars).
mardia_test <- function(X) {
  X <- as.matrix(X); n <- nrow(X); p <- ncol(X)
  Xc <- scale(X, center = TRUE, scale = FALSE)
  S <- cov(X) * (n - 1) / n
  D <- Xc %*% solve(S) %*% t(Xc)

  b1p <- sum(D^3) / n^2
  b2p <- sum(diag(D)^2) / n
  k1 <- n * b1p / 6
  df1 <- p * (p + 1) * (p + 2) / 6
  p1 <- pchisq(k1, df1, lower.tail = FALSE)
  z2 <- (b2p - p * (p + 2)) / sqrt(8 * p * (p + 2) / n)
  p2 <- 2 * pnorm(-abs(z2))

  data.frame(
    Test = c("Mardia skewness", "Mardia kurtosis"),
    Statistic = round(c(k1, z2), 3),
    df = c(df1, NA),
    p_value = signif(c(p1, p2), 3),
    Reject_H0 = ifelse(c(p1, p2) < 0.05, "Yes", "No"),
    stringsAsFactors = FALSE
  )
}

kable(mardia_test(X1), booktabs = TRUE,
      caption = "Mardia test of multivariate normality, L. torrens.") %>%
  kable_styling(latex_options = "hold_position")
kable(mardia_test(X2), booktabs = TRUE,
      caption = "Mardia test of multivariate normality, L. carteri.") %>%
  kable_styling(latex_options = "hold_position")
chiQQ <- function(M, ttl) {
  d2 <- mahalanobis(M, colMeans(M), cov(M))
  qchi <- qchisq(ppoints(nrow(M)), df = ncol(M))
  plot(qchi, sort(d2), pch = 20, main = ttl,
       xlab = expression(chi[7]^2 ~ "quantile"),
       ylab = "ordered d^2")
  abline(0, 1, col = "red", lwd = 1.5)
}
op <- par(mfrow = c(1, 2))

```

```

chiQQ(X1, "L. torrens"); chiQQ(X2, "L. carteri"); par(op)
cutoff <- qchisq(0.975, df = p) # ~ 16.01
d2_1 <- mahalanobis(X1, colMeans(X1), cov(X1))
d2_2 <- mahalanobis(X2, colMeans(X2), cov(X2))

out_tbl <- data.frame(
  Species = c(rep("L.torrens", n1), rep("L.carteri", n2)),
  Index = c(seq_len(n1), seq_len(n2)),
  D2 = c(d2_1, d2_2),
  Outlier = c(d2_1, d2_2) > cutoff
)
kable(subset(out_tbl, Outlier),
      digits = 2, booktabs = TRUE,
      caption = sprintf("Cases with  $d^2 > 0.2f$ .", cutoff)) %>%
  kable_styling(latex_options = "hold_position")
lambda <- sapply(1:p, function(k) {
  bc <- MASS::boxcox(X[, k] ~ 1, plotit = FALSE)
  bc$x[which.max(bc$y)]
})
names(lambda) <- colnames(X)
kable(t(round(lambda, 2)), booktabs = TRUE,
      caption = "Optimal Box-Cox lambdas (per variable).") %>%
  kable_styling(latex_options = "hold_position")
# Box-Cox helper: log when lambda is (numerically) zero.
bc_apply <- function(x, lam) {
  if (abs(lam) < 1e-6) log(x) else (x^lam - 1) / lam
}

Xbc <- sapply(1:p, function(k) bc_apply(X[, k], lambda[k]))
colnames(Xbc) <- colnames(X)
X1bc <- Xbc[g == "L.torrens", ]
X2bc <- Xbc[g == "L.carteri", ]

kable(mardia_test(X1bc), booktabs = TRUE,
      caption = "Mardia test after variable-specific Box-Cox, L. torrens.") %>%
  kable_styling(latex_options = "hold_position")
kable(mardia_test(X2bc), booktabs = TRUE,
      caption = "Mardia test after variable-specific Box-Cox, L. carteri.") %>%
  kable_styling(latex_options = "hold_position")
op <- par(mfrow = c(1, 2))
chiQQ(X1bc, "Box-Cox L. torrens")
chiQQ(X2bc, "Box-Cox L. carteri")
par(op)
# Re-check Mahalanobis outliers on the transformed scale
d2_1bc <- mahalanobis(X1bc, colMeans(X1bc), cov(X1bc))
d2_2bc <- mahalanobis(X2bc, colMeans(X2bc), cov(X2bc))
out_bc <- data.frame(

```

```

Species = c(rep("L.torrens", n1), rep("L.carteri", n2)),
Index   = c(seq_len(n1), seq_len(n2)),
D2      = c(d2_1bc, d2_2bc),
Outlier = c(d2_1bc, d2_2bc) > cutoff
)
kable(subset(out_bc, Outlier), digits = 2, booktabs = TRUE,
      caption = sprintf(
        "Cases with  $d^2 > %.2f$  after Box-Cox transformation.", cutoff)) %>%
  kable_styling(latex_options = "hold_position")
# Indices of multivariate outliers on the raw scale
out1_raw <- which(d2_1 > cutoff)
out2_raw <- which(d2_2 > cutoff)

X1c <- X1[setdiff(seq_len(n1), out1_raw), , drop = FALSE]
X2c <- X2[setdiff(seq_len(n2), out2_raw), , drop = FALSE]
Xc  <- rbind(X1c, X2c)
gc  <- factor(c(rep("L.torrens", nrow(X1c)),
                rep("L.carteri", nrow(X2c))),
              levels = c("L.torrens", "L.carteri"))

# Re-estimate Box-Cox lambdas on the cleaned data
lambda_c <- sapply(1:p, function(k) {
  bc <- MASS::boxcox(Xc[, k] ~ 1, plotit = FALSE)
  bc$x[which.max(bc$y)]
})
names(lambda_c) <- colnames(X)

Xbc_c <- sapply(1:p, function(k) bc_apply(Xc[, k], lambda_c[k]))
colnames(Xbc_c) <- colnames(X)
X1bc_c <- Xbc_c[gc == "L.torrens", ]
X2bc_c <- Xbc_c[gc == "L.carteri", ]
kable(rbind(`Full sample` = round(lambda, 2),
            `Outliers removed` = round(lambda_c, 2)),
      booktabs = TRUE,
      caption = "Optimal Box-Cox lambdas, before vs. after outlier removal.") %>%
  kable_styling(latex_options = "hold_position")
kable(mardia_test(X1bc_c), booktabs = TRUE,
      caption = sprintf(
        "Mardia test after Box-Cox + outlier removal, L. torrens (n = %d).",
        nrow(X1c))) %>%
  kable_styling(latex_options = "hold_position")
kable(mardia_test(X2bc_c), booktabs = TRUE,
      caption = sprintf(
        "Mardia test after Box-Cox + outlier removal, L. carteri (n = %d).",
        nrow(X2c))) %>%
  kable_styling(latex_options = "hold_position")
op <- par(mfrow = c(1, 2))

```



```

chiQQ(X1bc_c, "Cleaned L. torrens")
chiQQ(X2bc_c, "Cleaned L. carteri")
par(op)
bM_raw    <- biotools::boxM(X,      g)
bM_bc     <- biotools::boxM(Xbc,    g)
bM_clean  <- biotools::boxM(Xbc_c,  gc)
list(raw = bM_raw, box_cox = bM_bc, box_cox_clean = bM_clean)
ht_raw    <- ICSNP::HotellingsT2(X1,      X2)
ht_bc     <- ICSNP::HotellingsT2(X1bc,    X2bc)
ht_clean  <- ICSNP::HotellingsT2(X1bc_c,  X2bc_c)
ht_raw; ht_bc; ht_clean
simul_ci <- function(A, B, alpha = 0.05) {
  n1 <- nrow(A); n2 <- nrow(B); p <- ncol(A)
  xb1 <- colMeans(A); xb2 <- colMeans(B)
  Sp  <- ((n1 - 1) * cov(A) + (n2 - 1) * cov(B)) / (n1 + n2 - 2)

  c2 <- (n1 + n2 - 2) * p / (n1 + n2 - p - 1) *
    qf(1 - alpha, p, n1 + n2 - p - 1)
  tbf <- qt(1 - alpha / (2 * p), n1 + n2 - 2)

  diff <- xb1 - xb2
  se   <- sqrt(diag(Sp) * (1 / n1 + 1 / n2))

  data.frame(
    Variable = colnames(A),
    Diff      = diff,
    T2_Lo     = diff - sqrt(c2) * se,
    T2_Hi     = diff + sqrt(c2) * se,
    BF_Lo     = diff - tbf      * se,
    BF_Hi     = diff + tbf      * se
  )
}

CI_raw    <- simul_ci(X1,      X2)
CI_bc     <- simul_ci(X1bc,    X2bc)
CI_clean  <- simul_ci(X1bc_c,  X2bc_c)
kable(CI_raw, digits = 3, booktabs = TRUE,
      caption = "Simultaneous 95\\% CIs for mean differences (raw scale).") %>%
  kable_styling(latex_options = "hold_position")
kable(CI_bc, digits = 3, booktabs = TRUE,
      caption = "Simultaneous 95\\% CIs for mean differences (Box-Cox scale).") %>%
  kable_styling(latex_options = "hold_position")
kable(CI_clean, digits = 3, booktabs = TRUE,
      caption = "Simultaneous 95\\% CIs (Box-Cox + outliers removed).") %>%
  kable_styling(latex_options = "hold_position")
plt <- CI_raw %>%
  mutate(Variable = factor(Variable, levels = colnames(X)))

```

```
ggplot(plt, aes(Variable, Diff)) +
  geom_point(size = 2) +
  geom_errorbar(aes(ymin = BF_Lo, ymax = BF_Hi), width = 0.15) +
  geom_hline(yintercept = 0, linetype = 2, colour = "red") +
  coord_flip() + theme_bw() +
  labs(y = "Mean(L.torrens) - Mean(L.carteri)", x = NULL)
```

11 Appendix B: Session information

```
sessionInfo()
```

```
## R version 4.6.0 (2026-04-24)
## Platform: aarch64-apple-darwin23
## Running under: macOS Sequoia 15.5
##
## Matrix products: default
## BLAS: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib; LAPACK
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Los_Angeles
## tzcode source: internal
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
## other attached packages:
## [1] kableExtra_1.4.0 knitr_1.51 ICSNP_1.1-2 ICS_1.4-2
## [5] mvtnorm_1.3-7 biotools_4.3 MASS_7.3-65 MVN_6.3
## [9] GGally_2.4.0 ggplot2_4.0.3 tidyr_1.3.2 dplyr_1.2.1
## [13] readxl_1.4.5
##
## loaded via a namespace (and not attached):
## [1] tidyselect_1.2.1 viridisLite_0.4.3 farver_2.1.2 viridis_0.6.5
## [5] S7_0.2.2 fastmap_1.2.0 lazyeval_0.2.3 digest_0.6.39
## [9] rpart_4.1.27 lifecycle_1.0.5 survival_3.8-6 magrittr_2.0.5
## [13] compiler_4.6.0 rlang_1.2.0 tools_4.6.0 yaml_2.3.12
## [17] data.table_1.18.4 labeling_0.4.3 htmlwidgets_1.6.4 xml2_1.5.2
## [21] RColorBrewer_1.1-3 abind_1.4-8 withr_3.0.2 purrr_1.2.2
## [25] nnet_7.3-20 grid_4.6.0 jomo_2.7-6 mice_3.19.0
## [29] scales_1.4.0 iterators_1.0.14 tinytex_0.59 survey_4.5
## [33] cli_3.6.6 rmarkdown_2.31 reformulas_0.4.4 generics_0.1.4
## [37] otel_0.2.0 rstudioapi_0.18.0 httr_1.4.8 DBI_1.3.0
## [41] minqa_1.2.8 energy_1.7-12 stringr_1.6.0 splines_4.6.0
```

```
## [45] cellranger_1.1.0    mitools_2.4          vctr_0.7.3           boot_1.3-32
## [49] glmnet_5.0          Matrix_1.7-5         jsonlite_2.0.0       carData_3.0-6
## [53] car_3.1-5           mitml_0.4-5          Formula_1.2-5        systemfonts_1.3.2
## [57] nortest_1.0-4       foreach_1.5.2        plotly_4.12.0        glue_1.8.1
## [61] nloptr_2.2.1        pan_1.9              ggstats_0.13.0       codetools_0.2-20
## [65] stringi_1.8.7       shape_1.4.6.1        gtable_0.3.6         lme4_2.0-1
## [69] gsl_2.1-9           tibble_3.3.1         pillar_1.11.1        htmltools_0.5.9
## [73] R6_2.6.1            textshaping_1.0.5    Rdpack_2.6.6         evaluate_1.0.5
## [77] lattice_0.22-9      rbibutils_2.4.1      moments_0.14.1       backports_1.5.1
## [81] broom_1.0.12        Rcpp_1.1.1-1.1      svglite_2.2.2        gridExtra_2.3
## [85] nlme_3.1-169        xfun_0.57            pkgconfig_2.0.3
```

12 Appendix C: Reproducibility notes

- Place `flies.xls` in the same directory as this `.Rmd` file.
- Knit with: `rmarkdown::render("Midterm_Report.Rmd")`.
- PDF output requires a working LaTeX installation (e.g. `tinytex::install_tinytex()`).
- All random operations are seeded by `set.seed(2025)` at the top of the document, although none of the analyses in this report use randomness.